Retention-constrained class expansion with reference-anchored candidate tables

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Abstract

Reference-anchored candidate tables reuse a trained classifier for novel recognition under an original-class accuracy requirement. Fixed base references normalize learned, semantic and prototype evidence while preserving existing scores; a separate threshold controls novel admission. Matched-retention frontiers and paired arrival sequences quantify expansion capacity and classifier value. Across three datasets and two frozen CLIP encoders, independent protection-risk calibration with up to 40 images per base class certifies final-session expansion in all ten partitions per condition, with every 95% test-retention target met. ViT-L/14 novel accuracy reaches 72.76%, 88.51% and 80.74% on CIFAR-100, Pets and ImageNet-R. Retaining the classifier adds 0.71 and 0.90 percentage points over dense prototypes in ImageNet-R average incremental accuracy with ViT-L/14 and ViT-B/32.

Keywords: Class expansion, Retention constraints, Incremental learning, Evidence fusion, Vision–language models, Long-tailed recognition

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1. Introduction

An expanding recognition system already has a classifier whose decision boundaries reflect task-specific supervision. New classes arrive with names and a few examples. The central question is how much new-class recognition that existing discrimination can support while maintaining a specified accuracy on the original classes. A second question follows across repeated arrivals: how much predictive value is gained by keeping the trained classifier rather than replacing it with prototypes or recalibrated semantic evidence?

We address these questions with *reference-anchored candidate tables*. The table retains learned base columns and appends semantic anchors and support prototypes for arriving classes. Each evidence branch is normalized against fixed base references, so adding candidates leaves existing scores unchanged. An independent admission offset controls competition between base and novel classes. This separates the discrimination already learned from the decision to admit new alternatives.

The contribution is the connection between retained discrimination and class-expansion capacity: a persistent score frame makes the original classifier reusable, a common base-accuracy constraint makes its admission value comparable, and paired arrival sequences measure its continuing benefit. Independent protection-risk calibration adds a confidence-qualified admission rule and an explicit class-coverage requirement. On ImageNet-R, the gain over dense prototypes persists across two frozen encoders. With sufficient independent calibration coverage, the tables certify final-session expansion on every evaluated partition of all three datasets and both encoders.

2. Related work

Long-tailed recognition addresses unequal class supervision through balanced fitting and losses [1, 2, 3]. CLIP supplies semantic image–text evidence [4], while prompt adaptation [5, 6], prototypes [7, 8] and semantic caches [9] adapt pre-trained representations to labeled data. Our setting combines these evidence types according to their availability at class arrival.

Class-incremental and few-shot incremental learning study retention across successive tasks [10, 11]. SimpleCIL [12] and RanPAC [13] show the value of pre-trained prototypes and analytic readouts. KBS approaches develop multi-feature similarity supplementation [14], sharpness-aware prototype calibration [15], adaptive exploration [16], representation-enhanced analytic learning [17], and multimodal prompt fusion [18].

Recent work aligns task-specific expert inference in MoTE [19], separates expanding feature spaces through orthogonal constraints [20], and learns invariant representations through counterfactual distribution intervention [21]. Our contribution concerns the admission capacity supplied by an existing classifier under a common base-accuracy requirement. CLIP-guided discovery [22] additionally uses unlabeled novel data, whereas our arriving classes provide names and labeled support. FeNeC [23] supplies a clustered-representative readout for controlled frozen-feature comparisons.

BiMC [24] and HyCal [25] calibrate multimodal or heterogeneous recognition metrics. Calibrated seen/unseen recognition [26] studies the related operating trade-off, and Learn then Test [27] supplies risk-controlled threshold selection. We focus on the retained classifier as a source of admission capacity: its contribution is measured at a common base requirement, over successive

arrivals, and under a separately calibrated protection policy.

3. From retained discrimination to novel admission

3.1. Task and evidence

Base classes $\mathcal{B}$ have labeled, potentially imbalanced fitting data. Arriving classes $\mathcal{N}$ provide a name and k labeled support images per class. Prediction selects from $\mathcal{B} \cup \mathcal{N}$ without a group oracle. A frozen encoder produces a normalized feature $v = f(x)$; normalized text anchors t_c average eight fixed prompts. Learned base scores are $q_L(x, b) = v^\top w_b + b_b$, fitted by inverse-frequency-weighted ridge regression. Novel prototypes are normalized support means p_n, giving $q_P(x, n) = v^\top p_n$; semantic scores are $q_T(x, c) = v^\top t_c$.

For each evidence source e, a fixed base reference bank $\mathcal{R}_e$ defines

$$\mu_e(x) = |\mathcal{R}_e|^{-1} \sum_{b \in \mathcal{R}_e} q_e(x, b), \tag{1}$$

$$\sigma_e^2(x) = |\mathcal{R}_e|^{-1} \sum_{b \in \mathcal{R}_e} (q_e(x, b) - \mu_e(x))^2. \tag{2}$$

We use base text anchors, learned columns and visual means as the text, learned and prototype references. Normalized evidence is

$$\widehat{q}_e(x, c) = \frac{q_e(x, c) - \mu_e(x)}{\max(\sigma_e(x), 10^{-6})}. \tag{3}$$

The statistics adapt to each query while remaining independent of the active novel vocabulary. Fusion uses the evidence available to each row:

$$s(x, b) = \widehat{q}_T(x, b) + a\widehat{q}_L(x, b), \qquad b \in \mathcal{B}, \tag{4}$$

$$s(x, n) = \widehat{q}_T(x, n) + u\widehat{q}_P(x, n) + d, \qquad n \in \mathcal{N}. \tag{5}$$

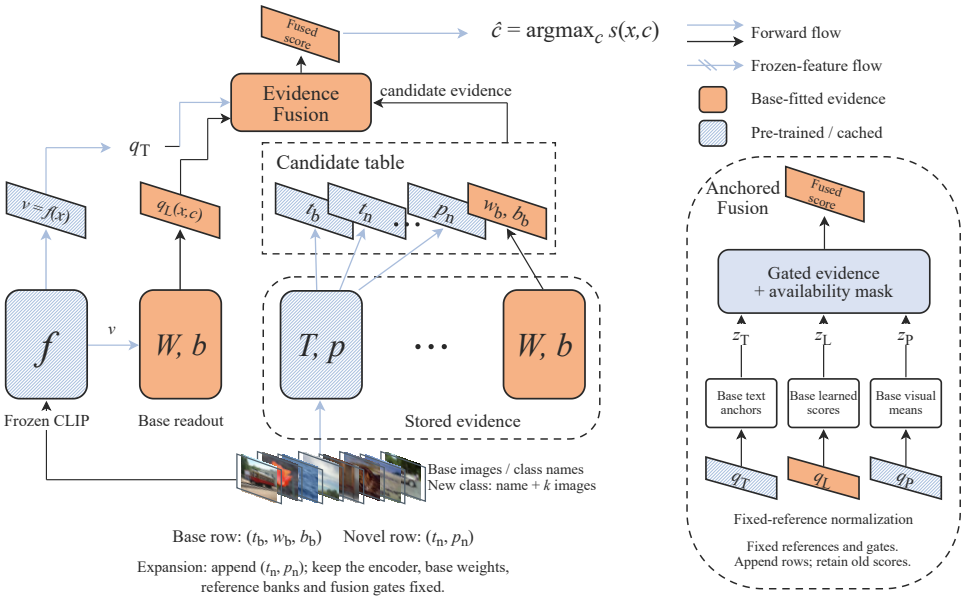


Figure 1: Reference-anchored candidate tables. Fixed base references set each branch's score frame; learned base columns remain stored as novel text anchors and support prototypes are appended. The separate novel offset controls admission. Orange marks base-fitted evidence; blue hatching marks pre-trained or cached evidence.

The coefficients a, u combine evidence; d controls novel admission (Fig. 1). Candidate-wise normalization instead changes the branch scales when new rows arrive, which can alter existing fused decisions. A numerical example appears in Supplementary Section K.

Proposition 1. *With fixed input, encoder, existing row evidence, reference banks, coefficients and tie-breaking, appending rows preserves every existing score and its within-vocabulary ordering.*

All terms of an existing row remain fixed. This structural property also holds for raw and fixed-temperature controls; their recognition comparisons

isolate reference normalization. Appended rows can still win the expanded prediction. Supplementary Section E gives the proof and prediction consequence. Each new row costs $O(D)$ storage and $O(kD)$ prototype construction after encoding; scoring K classes costs $O(KD)$.

3.2. Calibration and expansion capacity

A base-only pseudo-expansion episode splits base classes into pseudo-base and pseudo-novel groups. A classifier fitted on pseudo-base classes is combined with names and reserved support for pseudo-novel classes. Held-out base queries select a, u, d by harmonic macro accuracy,

$$H = 2A_B A_N/(A_B + A_N), \tag{6}$$

with zero assigned when both accuracies vanish. The coefficients then remain fixed across actual arrivals. Actual novel query labels and test labels are excluded from this selection; grids and split seeds are in Supplementary Section K.

Let $A_{B,\mathrm{ref}}^{\mathrm{closed}}$ denote the retained table's closed-base macro accuracy. At retention fraction r, every readout receives the same absolute base target. Its empirical expansion capacity is

$$N_r = \max_d A_N(d) \quad \text{subject to} \quad A_B(d) \geq r A_{B,\mathrm{ref}}^{\mathrm{closed}}. \tag{7}$$

A test-label sweep characterizes this descriptive frontier; an operational threshold uses calibration data. The exact margin envelope is derived in Supplementary Section D. Reporting these two evaluations separately distinguishes attainable capacity from selected-policy behavior.

3.3. Independent protection-risk calibration

Let R indicate correct prediction by the retained closed-base reference and C_d correct prediction by the expanded readout. Define

$$L_d = R(1 - C_d) + (1 - r)(1 - R). \tag{8}$$

Writing $D = \Pr(R = 1, C_d = 0)$ gives $\mathbb{E}L_d = D + (1 - r)(1 - A_{B,\mathrm{ref}})$. Thus $\mathbb{E}L_d \leq 1 - r$ implies $A_B(d) \geq rA_{B,\mathrm{ref}}$. This common sufficient criterion conservatively ignores replacement readouts' recovery of reference errors.

Fitting and tuning data fix the score models and ordered candidate thresholds. Independent, class-stratified calibration observations test thresholds from conservative to permissive using weighted bounded-loss p-values and fixed-sequence testing [27]. Testing stops at the first failure. Allocating error probability $\delta/5$ per arrival controls population protection across five prespecified sessions for each method, order and partition under independent class-conditional sampling. It does not guarantee arbitrary distribution shifts or every finite test outcome. If no finite threshold passes, the policy returns the reference classifier with zero novel admission. Supplementary Section J provides the bound, assumptions and full selection rule.

4. Experiments

4.1. Protocol and comparisons

We evaluate CIFAR-100 [28], Oxford-IIIT Pets [29] and ImageNet-R [30], using frozen OpenAI CLIP ViT-L/14 and ViT-B/32 features, five-shot support, ten frequency-stratified class partitions and two five-session arrival orders. Base/novel counts are 40/40, 15/15 and 100/100; each session adds eight,

three or 20 classes. Base pools follow long-tailed sampling. ImageNet-R uses task-specific splits after removing byte-identical duplicates. Novelty is relative to task fitting, with CLIP's pre-training exposure retained. Detailed pools, preprocessing and seeds are in Supplementary Sections C, I and K.

Every comparator shares its encoder, partition and support. We compare fixed-gate classifier removal, independently recalibrated removal, dense anchored text–prototype fusion, summed/class-mean Tip-Adapter caches, and raw/fixed-temperature retained fusion. The latter uses a single base-validation scale per branch. All mixtures use base-only selection; FeNeC and earlier three-partition auxiliary studies are retained in Supplementary Sections A–H.

Orders are averaged within a partition. Means and standard deviations describe ten class assignments on fixed images. Paired differences use identical cells; constrained comparisons use jointly feasible sets. Conditional paired 95% bootstrap intervals resample partition means 20,000 times. Average incremental accuracy is the mean macro accuracy over five arrivals. These intervals characterize class-assignment variation, conditional on the fixed data and encoders.

4.2. Novel recognition at a shared base requirement

At the descriptive 95% retention budget, the ImageNet-R L/14 table reaches 84.31% novel accuracy at 86.42% base accuracy. Figure 2 compares novel recognition under shared base requirements across encoders and datasets. A more discriminative base readout can tolerate greater novel competition while meeting the same target, turning retained discrimination into admission capacity.

Reference normalization also improves the sequential comparison against

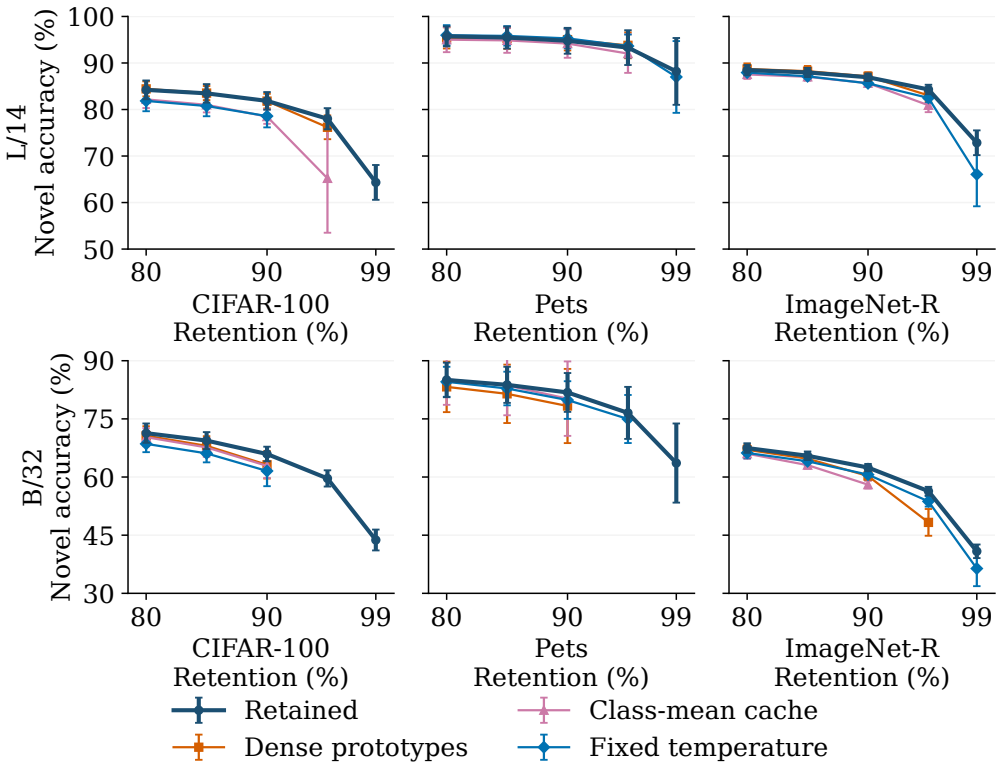


Figure 2: Novel recognition at shared base-retention requirements after the fifth arrival. Each readout must attain the stated fraction of the retained reference's closed-base accuracy within the same partition and encoder. Means and sample standard deviations cover ten partitions after averaging two arrival orders; points require all ten partitions to be feasible. Vertical ranges are 50–100% (L/14) and 30–90% (B/32); error bars stop at panel limits. Test-label threshold sweeps characterize descriptive capacity. Partial feasible sets and exact counts are reported in Supplementary Section I.

fixed-temperature fusion on ImageNet-R by 0.94 [0.74, 1.12] percentage points with L/14. Both preserve existing scores, so this comparison measures the value of query-dependent branch scaling. Full normalization controls are in Supplementary Section I.

4.3. The classifier's value across arrivals

Table 1 quantifies retained-minus-comparator average incremental accuracy with fixed base-only selected gates and offsets. On ImageNet-R, retaining the classifier adds 0.71 [0.46, 0.92] points over dense prototypes with L/14 and 0.90 [0.69, 1.11] with B/32. Corresponding gains over class-mean caches are 1.42 [1.15, 1.62] and 1.71 [1.44, 1.98] points. The result persists across both encoders and after calibration is restored to the classifier-free alternative.

Table 1: The retained classifier's value across five arrivals. Cells show paired gains in average incremental accuracy (percentage points), with conditional paired 95% bootstrap intervals below. Positive values favor retention; bold estimates have intervals excluding zero. Two orders are averaged within each of ten partitions. Full final-session differences are in Supplementary Section I.

Comparator	Retained-minus-comparator gain (pp) ↑		
	CIFAR-100	Pets	ImageNet-R
CLIP ViT-L/14			
Recalibrated removal	**+3.00** [+2.23, +3.75]	-0.14 [-1.53, +0.86]	**+2.30** [+1.95, +2.59]
Dense prototypes	+0.15 [-0.46, +0.64]	-0.98 [-2.52, +0.22]	**+0.71** [+0.46, +0.92]
Class-mean cache	**+1.44** [+0.68, +2.10]	-0.06 [-1.76, +1.41]	**+1.42** [+1.15, +1.62]
CLIP ViT-B/32			
Recalibrated removal	**+2.38** [+1.44, +3.34]	-0.44 [-1.64, +0.81]	**+3.07** [+2.82, +3.34]
Dense prototypes	+0.86 [-0.15, +1.76]	-0.39 [-1.85, +1.02]	**+0.90** [+0.69, +1.11]
Class-mean cache	+0.96 [-0.04, +1.84]	-0.74 [-1.62, +0.12]	**+1.71** [+1.44, +1.98]

CIFAR also shows gains over recalibrated removal with both encoders; its dense-prototype intervals span zero. Pets' average incremental intervals span zero for these comparators. The gain over prototype replacement is clearest on ImageNet-R.

4.4. Turning capacity into calibrated admission

An empirical calibration threshold selects the largest offset satisfying r times the reference's calibration accuracy. We separately evaluate whether

test accuracy meets r times its test reference. In the original L/14 pools, the retained 95% policy is validation-feasible in all ten partitions but meets the test target in 7/10 CIFAR, 1/10 Pets and 2/10 ImageNet-R partitions. Supplementary Section I reports all methods; Section H varies threshold-label budgets.

For independent risk calibration, the original fitting/validation pool is repartitioned into approximately 40% fitting, 20% tuning and 40% calibration observations per class. All readouts are refitted and tuned without calibration access (Supplementary Section J). We then freeze the models, support, test archives and 101 tuning-margin threshold candidates. Class-wise calibration caps of 2, 5, 10, 20 and 40 use nested subsets, supplementing the independent pool with unused training images while excluding fitting, tuning, support and test images. Every method receives identical subsets. Three subset seeds are averaged within each partition; the full cap-40 pool is counted once. We use $r = 0.95$ and $\delta = 0.05$. These benchmarks previously informed development; this is a revised benchmark evaluation.

Class coverage determines whether certification is attainable. For macro weights $w_i = 1/(Bn_c)$ and $b = \max_i w_i$, even zero observed loss requires $1/b \geq \log(\delta/5)/\log r = 89.78$ to pass the weighted bound. The original CIFAR and Pets allocations have effective counts 80 and 45–60, respectively. Increasing tail-class coverage makes useful certification possible (Fig. 3); total query counts and the derivation are in Supplementary Section L.

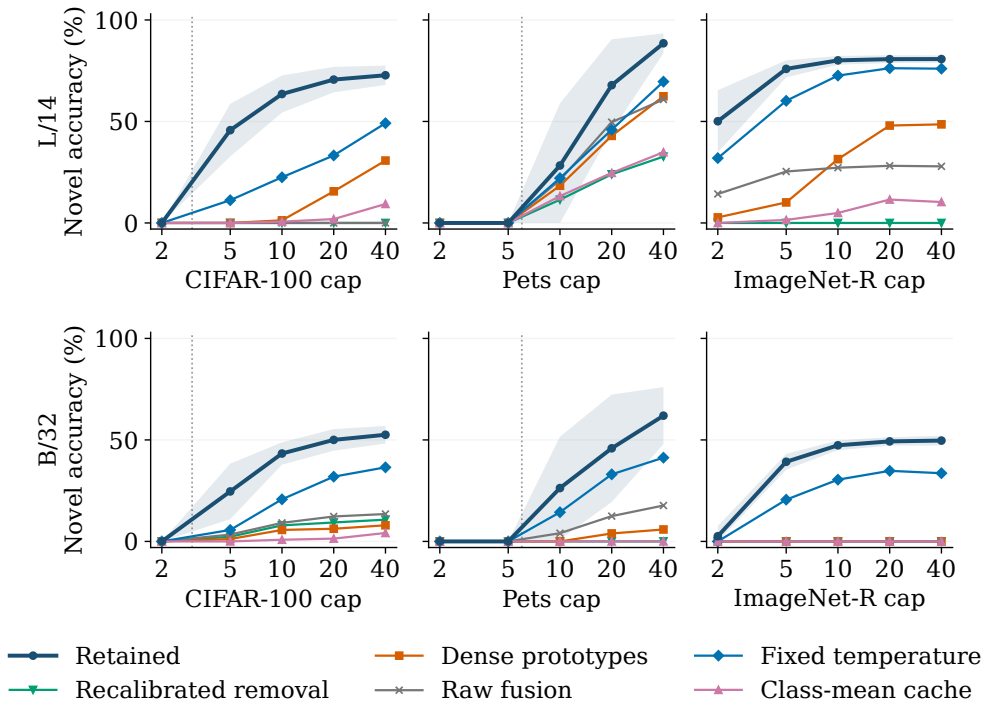


Figure 3: Novel accuracy after the fifth arrival in the first prespecified order, under independent 95% retention calibration with 95% five-arrival confidence. Means cover ten partitions, averaging three nested-subset seeds below cap 40 and the full set once at 40. Fallback contributes zero novel accuracy. Shading shows retained-table partition standard deviations, clipped to [0,100]. Grey lines mark necessary balanced class caps (CIFAR: 3; Pets: 6). ImageNet-R availability truncates the cap. All methods share the same calibration images; full counts and certification curves are in Supplementary Section L.

Table 2: Independently calibrated admission after the fifth arrival in the first prespecified order, with up to 40 images per base class, 95% retention and 95% five-session confidence. Queries are actual base-calibration counts; Finite counts certified expansion and Test met counts attained test targets. Base/Novel are test percentages, mean ± sample standard deviation over ten partitions, including fallback. All six readouts are reported in Supplementary Section L.

Dataset	Calibration	Admission		Test accuracy (%) ↑	
	Queries	Finite	Test met	Base	Novel
CLIP ViT-L/14					
CIFAR-100	1600	10/10	10/10	85.86 ± 1.19	72.76 ± 4.76
Pets	600	10/10	10/10	94.10 ± 2.27	88.51 ± 4.85
ImageNet-R	3636–3778	10/10	10/10	89.02 ± 0.51	80.74 ± 1.83
CLIP ViT-B/32					
CIFAR-100	1600	10/10	10/10	76.74 ± 1.73	52.54 ± 4.34
Pets	600	10/10	10/10	88.10 ± 3.41	61.94 ± 14.07
ImageNet-R	3636–3778	10/10	10/10	72.98 ± 1.21	49.67 ± 2.38

At cap 40, retained tables certify final-session expansion in all ten partitions of each dataset and encoder, with all test targets met (Table 2). L/14 novel accuracy is 72.76% on CIFAR, 88.51% on Pets and 80.74% on ImageNet-R. All 20 arrival paths per dataset/encoder meet every test target. Finite admission holds throughout all CIFAR and ImageNet-R paths and 17/20 L/14 and 16/20 B/32 Pets paths; the remaining Pets paths use an early-session fallback. Full comparisons distinguish policy gains, which include fallback, from accuracy differences on jointly admitted partitions. The common protection criterion

remains conservative for replacement models that recover reference errors.

5. Conclusion

Candidate tables turn retained discrimination into reusable expansion capacity. Fixed references preserve the existing score frame, while an admission threshold exposes the trade-off between original-class protection and novel recognition. Matched-retention and paired sequential comparisons quantify this value. Independent calibration with adequate class coverage supports protected admission across three datasets and two encoders, connecting what an existing classifier has learned to how much new recognition it can support.

CRediT authorship contribution statement

Junyi Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

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Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The experiments use public CIFAR-100, Oxford-IIIT Pets and ImageNet-R data and the publicly released CLIP ViT-L/14 and ViT-B/32 checkpoints. Code, source-hashed experimental records, prediction archives, manuscript sources, and table/figure generation scripts are available at `https://github.com/Cheerssom/Reference-Anchored-Candidate-Tables`. Dataset images, pretrained weights, and feature caches are obtained separately using the repository instructions.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex was used to assist with literature discovery, manuscript drafting and revision, experiment and verification scripts, and LaTeX and table preparation. The author is responsible for the accuracy, interpretation and originality of the submitted work and for reviewing and approving the final manuscript and supporting materials.

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